

Gut Microbiome Differences Between Two Dietary Groups

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Study Aim

The analysis focuses on whether gut microbial communities differ between two dietary groups: children consuming a Mediterranean diet and children consuming a plant-based rural diet described as vegetarian (sorghum, millet, and black-eyed peas). The central analytical question is whether these two dietary groups show detectable differences in gut microbial composition, within-sample diversity, between-sample community structure, and genus-level abundance.

Dataset Overview

The raw dataset (HTML output Section 3.1)

The dataset ([ENA Project: ERP000133](#)) includes 29 samples: 15 from the Mediterranean diet group and 14 from the vegetarian diet group. Raw sequencing depth varied across samples, ranging from 9,633 to 25,647 reads per sample, with a median of 15,583 and a mean of 16,210 reads. Total raw reads were broadly comparable between groups: 243,231 reads in the Mediterranean group and 226,864 reads in the vegetarian group, suggesting that the two groups are reasonably balanced in total sequencing effort.

Summary of quality scores and sequence truncation (HTML output Section 3, 4.1)

Read quality was assessed using DADA2 'plotQualityProfile()' output (Section 3.1.1). The average quality scores were generally around Q30-Q40 before 300 bp, indicating low expected base-call error in this region. After approximately 300 bp, quality scores declined to around Q10-Q20, indicating a clear deterioration in read quality. This pattern supports trimming or filtering reads around the 300 bp regions. The filtering strategy retained reads between 200 and 310 bp for two reasons: 1) most samples had median read lengths close to 300 bp; 2) to remove overly long low-quality tails.

After filtering and trimming (Section 3.2), reads.out ranged from 7,531 to 19,369 reads per sample, with a median of 11,649 and a mean of 12,456. The overall mean read-retention rate was 0.770, and the median retention rate was 0.775. Retention was slightly higher in the Mediterranean group, with a mean of 0.7866, compared with 0.7489 in the vegetarian group. This suggests that the vegetarian group may have had slightly more low-quality, short, or otherwise filtered reads, although both groups retained sufficient reads for downstream analysis.

The final non-chimeric read count represented approximately 60.0% of raw reads (Section 4.1.2). The chimera-removal step retained 80.98% of reads relative to the sequence table before chimera removal.

Analytical Workflow and Statistical Methods

The analysis used a standard DADA2 workflow. First, FASTQ files and participant data were loaded. The diet variable was recoded into 'mediterranean' and 'vegetarian', with the Mediterranean group set as the reference category. Then, quality-profile inspection, read-length summarization, filtering, and trimming are performed. Filtered reads were used to learn error rates, followed by dereplication and denoising to infer ASVs.

After ASV-table construction and chimera removal, 1,716 non-chimeric ASVs were retained. The post-chimera table contained no singletons, five doubletons, and a minimum ASV total abundance of 2. Taxonomy was assigned using the RDP release 19 training set, with classification reported to the genus level. A phylogenetic tree was built by aligning ASV sequences and constructing an initial neighbor-joining tree. The tree was then optimized under a GTR + Gamma + invariant-sites model using maximum likelihood. The ASV table, taxonomy table, tree, and participant metadata were integrated into a 'phyloseq' object.

Potential contaminant ASVs were checked by looking for *Chloroplast* and *Mitochondrial* classifications. None were reported. ASVs without Kingdom assignment were removed before downstream analysis, reducing the object from 1,716 to 1,714 ASVs. Taxonomic composition was summarized after transforming counts to relative abundance. Downstream analyses included phylum- and genus-level relative-abundance plots, alpha diversity, beta diversity, PERMANOVA, PERMDISP, and genus-level differential abundance analysis using ANCOM-BC2. Alpha diversity was evaluated using Observed richness, Shannon diversity, and Simpson diversity. Beta diversity was evaluated using Bray-Curtis distance and visualized with PCoA. Differential abundance analysis was performed with ANCOM-BC2 on genus-level count data. The model included Diet as the fixed effect, used BH correction for multiple testing.

Key Findings

Taxonomic composition (Section 7.1)

At the phylum level, the top eight phyla by mean relative abundance were plotted. *Bacteroidota* was the dominant phylum in the vegetarian group, followed by *Bacillota*. In contrast, *Bacillota* was the dominant phylum in the Mediterranean group, with *Bacteroidota* being the second most abundant phylum. At the genus level, the top 15 genera were plotted, unclassified taxa were labeled as Unclassified, and lower-abundance taxa were grouped as Other. At the genus level, *Segatella* was the most abundant genus in the vegetarian group, whereas *Faecalibacterium* was the most abundant genus in the Mediterranean group.

Alpha diversity (Section 7.2)

Differences between dietary groups were tested using Wilcoxon rank-sum tests, and p values across alpha-diversity metrics were adjusted with the Benjamini-Hochberg method. The HTML output reports meaningful group differences for all three metrics:

Observed richness $p = 0.025$, BH-adjusted $p = 0.025$; Shannon diversity $p = 0.002$, adjusted $p = 0.003$; and Simpson diversity $p = 0.001$, adjusted $p = 0.003$.

Beta diversity (Section 7.3)

The effect of dietary group on overall community composition was tested using PERMANOVA with 999 permutations. The PERMANOVA result was meaningful with an $R^2 = 0.09938$, $F = 2.9794$, $p = 0.001$. Because PERMANOVA can be influenced by differences in within-group dispersion, PERMDISP was also performed. PERMDISP results with an $F = 0.9959$ and $p = 0.305$, suggesting that the PERMANOVA result was not simply driven by one group being more dispersed.

Differential abundance analysis (Section 7.4)

Dorea, *Ruminococcus*, *Mediterraneibacter*, *Subdoligranulum*, and *Romboutsia* are identified to be associated with the two dietary groups. The abundance of *Dorea* was estimated to be 6.1-fold higher in the vegetarian group than in the mediterranean group (95% CI: 3.2 - 11.6). The abundance of *Subdoligranulum* was approximately 72% lower in vegetarians than in mediterranean (fold change = 0.28, 95% CI: 0.14 - 0.58).

Biological Interpretation

Long-term diet may shape the gut microbiome through differences in fermentable carbohydrates, dietary fiber, fat, and protein intake. The vegetarian group represents a rural, plant-based diet rich in grains and legumes, while the Mediterranean group represents a European childhood dietary context. The meaningful difference in Bray-Curtis PERMANOVA result indicates that the two groups differ not only in selected taxa but also in overall microbial community structure.

Study Limitations and Recommendations for Improving the Analysis

The major limitation is that dietary group is strongly confounded with geography, ethnicity/population background, living environment, and socioeconomic conditions. The bias that caused by not including additional variables can be reduced by including necessary explanatory variables into the model.

Another aspect that could be improved is the optimization of the phylogenetic tree. In this analysis, nearest-neighbor interchange (NNI) was used to reduce computational time. A more comprehensive approach could compare multiple nucleotide substitution models and optimization strategies. The best-fitting model could then be selected based on criteria such as AIC or BIC before final tree optimization.

The small sample size of only 29 samples also limited statistical power and the stability of the results. Although the alpha-diversity tests and PERMANOVA indicated meaningful differences between the two dietary groups, the estimated effect sizes and differential-abundance results may be unstable and should therefore be interpreted cautiously.